

Subject: Application for Senior Software Engineer (Automation Core) – Adrian Vrabie

Dear Roche Hiring Team,

I am writing to apply for the Senior Software Engineer position within the Automation Core team. My background combines a Master's in Mathematics with over a decade of engineering experience, primarily focused on building distributed systems that bridge complex domains with production-grade software.

I would like to highlight a few instances where I have taken complex concepts through to production, as I believe this experience mirrors the challenges of modernizing lab automation:

- **From Research to Global Scale (Comcast):** In 2018, I co-pioneered an SD-WAN product for Comcast. Starting from a specialized research paper, my team and I moved from initial concepts to a functional Proof of Concept and finally to a commercial product that is now utilized worldwide. This involved translating theoretical networking concepts into a scalable Java and Kubernetes architecture while collaborating closely with networking engineers in the USA to ensure the final product met rigorous field requirements.
- **End-to-End Delivery in Complex Migrations (UBS):** During my current tenure at UBS, I have focused on taking architectural concepts through to production. I led the development of the Facade service, which managed critical document ingestion during the Credit Suisse migration, and the Papdocs Generator. Both projects involved moving from initial requirements to POCs and finally to stable production deployments, ensuring that high-performance reactive microservices (Spring WebFlux) met the stringent security and performance standards of a global bank.
- **Analytical Foundations:** My MSc in Mathematics, specifically my work on Hidden Markov Models, provides the analytical foundation I use to solve non-trivial engineering problems. Whether it was optimizing algorithms at Comcast (reducing complexity from polynomial to linear) or leading strategy at MoldATSA, my focus has been on translating abstract needs into reliable, interconnected systems.

I am genuinely interested in how the Automation Core team integrates diverse hardware and software stacks to empower scientists. I would welcome the opportunity to discuss how my experience in delivering full-lifecycle solutions can support Roche's goals.

Thank you for your time and consideration.

Best regards,

Adrian Vrabie

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